

You run kubectl apply.

What happens next?

6 components. 1 path. Most engineers can't explain the full journey.



6 steps from YAML to running container.

01

API Server

`authn → authz → admission`



02

etcd

`persistent store · protobuf`



03

Deployment Controller

`watch → create ReplicaSet`



04

ReplicaSet Controller

`watch → create Pod objects`



05

Scheduler

`filter → score → bind nodeName`



06

Kubelet

`CRI → containerd → running ✓`



Your manifest passes 4 gates before storage.

```
kubectl apply -f deploy.yaml  
your YAML manifest
```



API Server

Authn

Authz

Mutating

Validating



admitted

etcd

```
/registry/deployments/ns/name → protobuf
```

Admission controllers can mutate fields, inject sidecars, or reject entirely — before anything is stored.



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Pods exist before any node knows.

Deployment Controller

watch → RS



creates ReplicaSet

ReplicaSet Controller

watch → Pods



creates 3 Pod objects

3× Pods in etcd

nodeName: ""

The empty **nodeName** is the key detail. Pods are real objects in etcd — they just don't have a home yet.



Scheduler scores. Kubelet executes.



scheduler → kubelet

```
# Scheduler watches unassigned pods  
watch: pods where spec.nodeName == ""  
  
# Filter → score → pick best node  
scheduler → bind:  
    spec.nodeName: "worker-2"  
  
# Kubelet on worker-2 picks it up  
kubelet → CRI → containerd → running ✓
```

No direct communication between scheduler and kubelet. Both talk to the **API server**. etcd is the single source of truth.



"Scheduler crashed mid-flight.
Are your pods gone?"

No.

Kubernetes is **level-triggered**, not edge-triggered.
Controllers reconcile **desired state vs actual state** —
not events.

Scheduler restarts → reads etcd → finds pods with
empty **nodeName** → assigns nodes. Nothing lost.



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